

Section 1: Metadata

to be filled by the student

1.1. Project Information to be filled by the student

Title: Astronomical Data Management System	
Section:	Instructor:

1.2. Student(s) Information

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Submission guideline: Save your project proposal as a pdf file and rename as Project Proposal_L1_ProposedTitle where L1 is to be replaced with your section

Section 2: The Project

to be filled by the student

2.1. Project Description: *Please provide a brief introduction of the project including its scope.*

The project titled **Astronomical Data Management System** is a desktop application aimed at managing astronomical data, including star positions, variable stars, celestial events, and astrological predictions (e.g., horoscopes, sun rising, moon rising). The system will allow users to browse, search, and explore stellar data, view celestial events, and access daily astrological insights.

The system will provide different user roles, such as administrators, power users, and general users, each with varying levels of access. The goal is to create an easy-to-use yet robust database management system that caters to both astronomers and astrology enthusiasts.

2.2 Functional Requirements

This section describes each function/feature provided by your system. These functions are logically grouped into modules based on their purposes. The users in your system must be categorized such as client, customer or administrator etc. These users will be accessing the database with the level of access that they are authorized with.

Sample functional Requirements:

Module 1: Registrations

- Function 1: Register an account

The system lets users register an account on the website as a client and as a customer.

– Function 1a: Register as a client The register form prompts the user to enter their details i.e. Name, Email, Password. The form is submitted and an unverified client account is created. The user receives a link on their email address which completes account verification.

– Function 1b: Register as a customer The register form prompts the user to enter their details i.e. Name, Email, Password. The form is submitted and an unverified customer account is created. The user receives a link on their email address which completes account verification.

Module 2: Buying and Selling

- Function 1: Only client type users can upload item(s) to sell .A registered client can upload an item to sell through the mobile application. The system prompts the client for item photos, item category and item details and the item is added to the item inventory.
- Function 2: Add item to cart. A certain button on the item page prompts the system to add the item to the user's cart. All consequent item(s) added without checkout, are added to the same cart unless cleared.
- Function 3: Clear item from cart .The system allows the user to remove a previously added item from the cart. If cart is empty, the checkout link is no longer accessible.

Module 1: Stellar Data Management

- **Function 1:** Add, edit, delete stellar data. This system allows admins to manage stellar data, including adding new star positions, editing existing data, and deleting outdated information.
- **Function 2:** Allows users to view stellar data for work or leisure. They can also search in the table using the Star Name and/or ID.

Fields include: Star ID, Name, Right Ascension (RA), Declination (Dec), Magnitude, Spectral Class, Distance, Proper Motion

Module 2: Variable Star Data

- **Function 1:** Add, edit, and delete variable stars. The system allows admins to manage variable stars, tracking changes in brightness over time.
- **Function 2:** Allows users to view variable stars data. They can also search in the table using the Star Name and/or Type.

Module 3: Celestial Event Management

- **Function 1:** View upcoming celestial events. Users can view details about upcoming events like meteor showers and eclipses, along with visibility information.

Module 4: Horoscope & Astrological Insights

- **Function 1:** View daily horoscope, sun rising, and moon rising times. All users can access their daily horoscope and view astrological data, including sun and moon rising times and predictions based on astrological charts.

CS355 – Database Project Proposal Form

2.3. Planned Schedule: *Kindly list the start/end dates and the timeline for the achievement of any intermediate milestones and the expected contribution to be made by the participant(s).*

Milestone	Task Description	Start Date	End Date	Responsible Party
1	Research and gather astronomical data	September 15	October 5	All
2	Database schema design	October 6	October 20	All
3	Implementing database and core features	October 21	November 5	All
4	GUI design and integration with PyQt6	November 6	November 20	All
5	Testing and final adjustments	-	-	All

Collaborative Approach: We will be working collaboratively throughout the project. Tasks will be distributed among team members based on expertise and learning objectives. As we progress, we will meet regularly to review each other's work, provide feedback, and make necessary adjustments to ensure the successful completion of the project.

This schedule allows for structured progression through the project, with overlapping dates providing flexibility for ongoing collaboration and revision as needed.

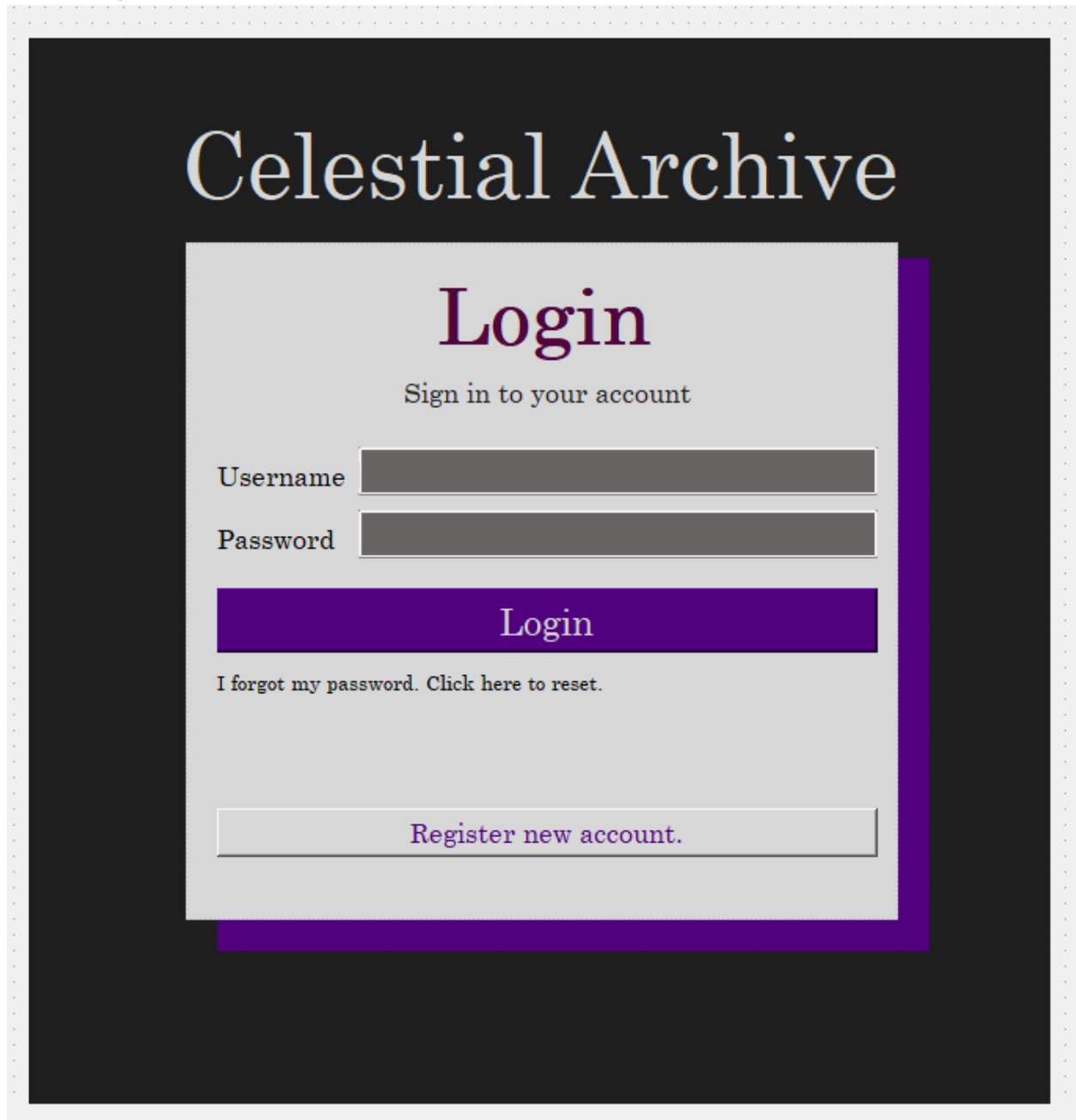
2.4. Technology Stack: *If you are utilizing any language or database besides PyQt and SQL Server, please complete this section; otherwise, leave it blank. Specify the programming language and database management system intended for constructing this application, as well as the application type (Desktop, Web, or Mobile).*

2.5. Screens: Provide images of all application screens, showcasing clear input and corresponding outputs. Ensure each image includes a concise caption explaining user action and expected/observed output. You can create these screens using Qt Designer.

Landing Page:



User Login:



The image shows a login form for 'Celestial Archive'. The form is centered on a dark background with a light gray border. It features a title 'Celestial Archive' in a large, serif font. Below it, the word 'Login' is written in a large, purple serif font. Underneath 'Login' is the text 'Sign in to your account' in a smaller, gray sans-serif font. There are two input fields: 'Username' and 'Password', both with gray borders and light gray text. Below the 'Password' field is a purple button with the text 'Login' in white. Under the button is a link that says 'I forgot my password. Click here to reset.' in a small, gray sans-serif font. At the bottom of the form is a light gray button with the text 'Register new account.' in a purple sans-serif font.

Celestial Archive

Login

Sign in to your account

Username

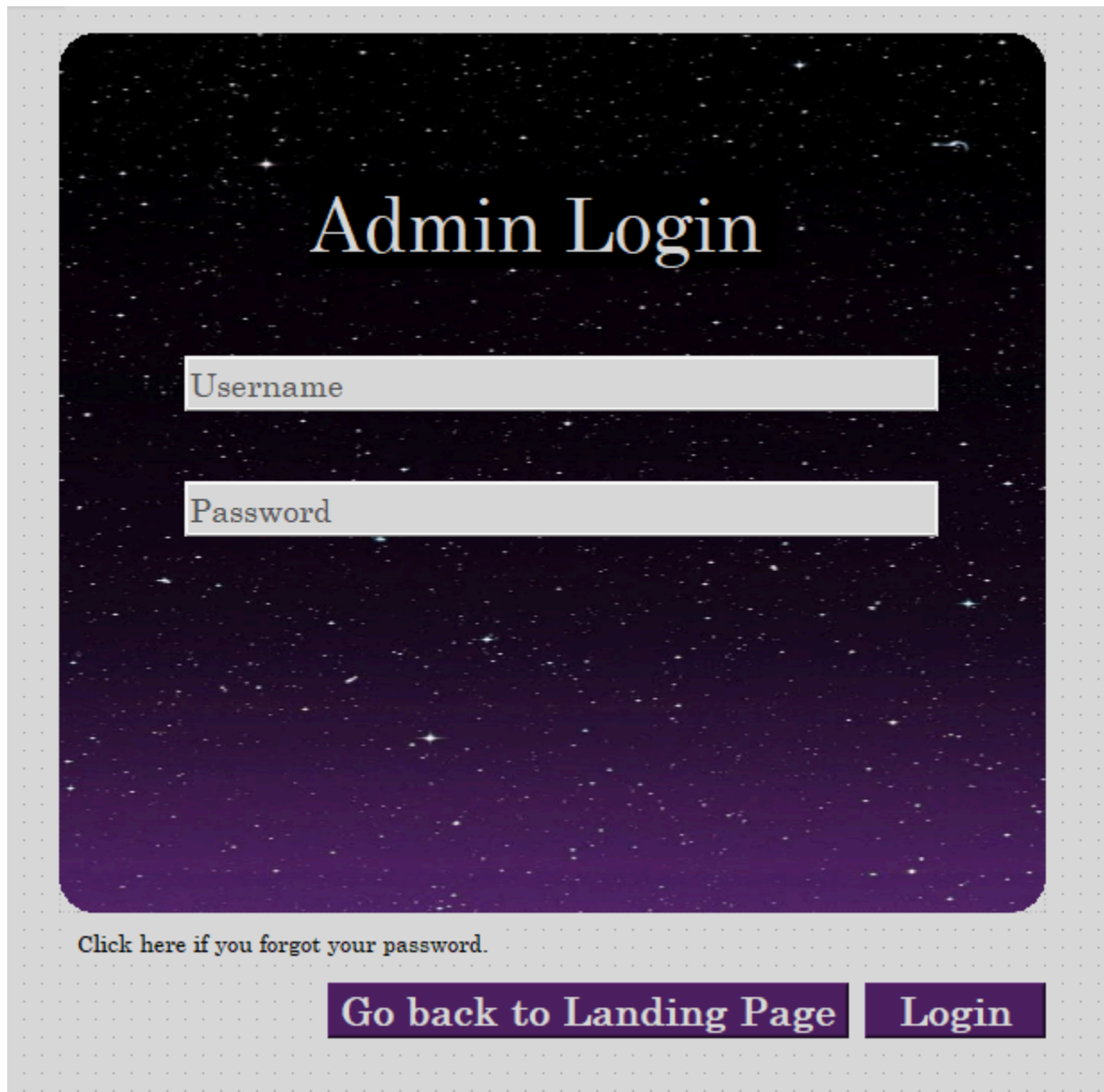
Password

Login

I forgot my password. Click here to reset.

Register new account.

Admin Login (for Admin):



The image shows a screenshot of an 'Admin Login' web form. The form is set against a dark, starry space background. It features two input fields: 'Username' and 'Password'. Below these fields is a link that says 'Click here if you forgot your password.' At the bottom of the form are two buttons: 'Go back to Landing Page' and 'Login'.

Admin Login

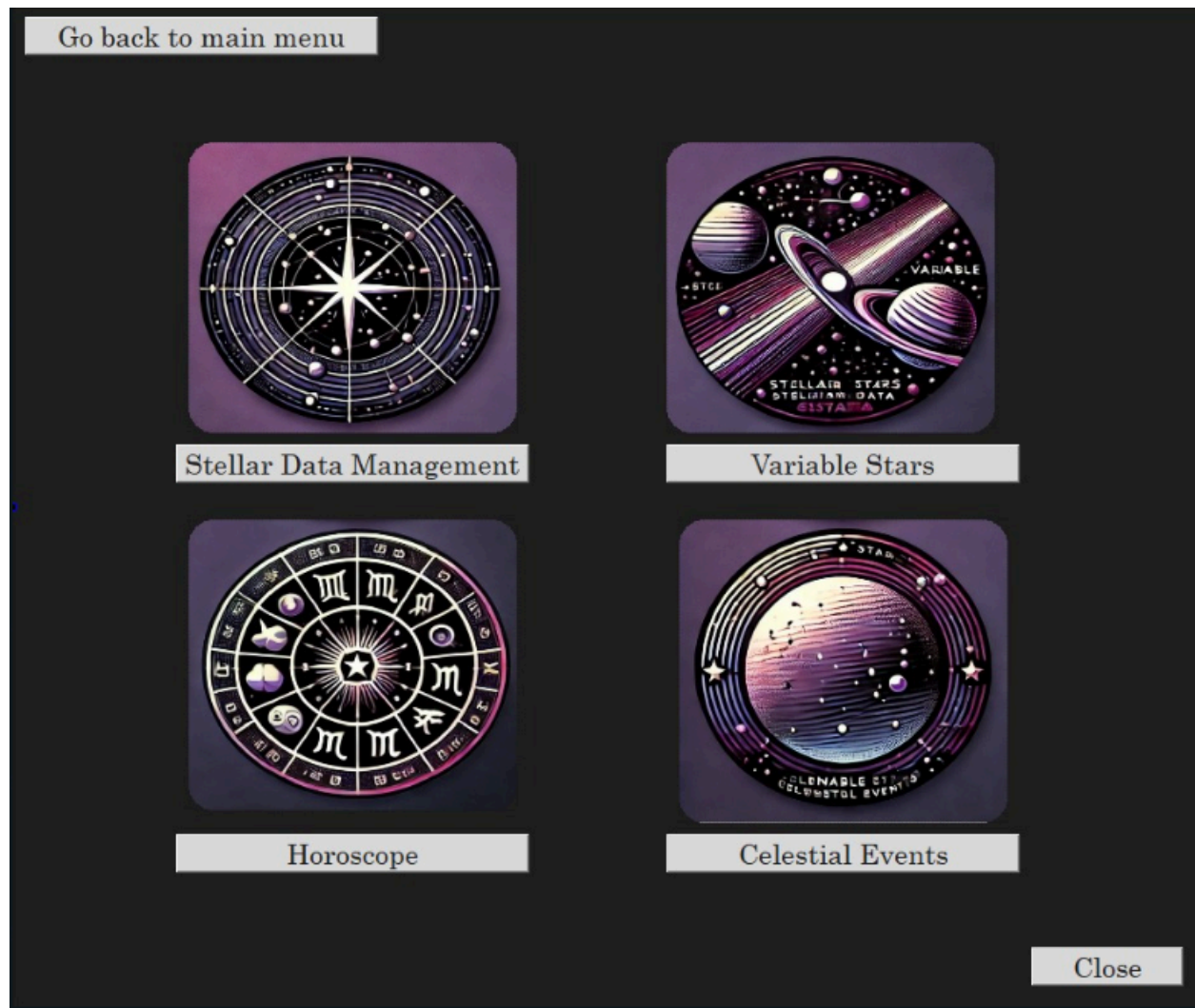
Username

Password

[Click here if you forgot your password.](#)

[Go back to Landing Page](#) [Login](#)

Menu (for Users and Admin):



Variable Stars (for Users):

Variable Stars

Star ID

Type

	ID	Max Magnitud	Min Magnitud	Type	Period	bservation ty
1						
2						
3						
4						
5						
6						
7						
8						

Horoscope Calculation Page (for Users):

HOROSCOPE

Enter your date and time of birth:

Enter your place of birth:

Calculate!



Horoscope main (for Users):

Star Sign

Name	
House	
Symbol	
Ruling Planet	

Sun Rising

Moon Rising



Daily Horoscope

This is placeholder text to check how it looks when a paragraph is typed in the text edit. Sagittarius, the ninth sign of the zodiac, is the home of the wanderers of the zodiac. It's not a mindless ramble for these folks, either. Sags are truth-seekers, and the best way for them to do this is to hit the road, talk to others and get some answers.

Knowledge is key to these folks since it fuels their broad-minded approach to life. Those born with a Sagittarius zodiac sign are keenly interested in philosophy and religion, and they find that these disciplines aid their internal quest. At the end of the day, what Sagittarius wants most is to know the meaning of life, and to accomplish this while feeling free and easy.

[Go back to main menu](#) [Close](#)

Stellar Data (for Users):

Stellar Data Management

Search

Name

Star ID

Search

	Star ID	Name	ht Ascension (	eclination(De	Magnitude	Spectral Class	Distance	Proper Motion
1								
2								
3								
4								
5								
6								
7								
8								

Go back to menu

Close

Celestial Event Viewer (for Users):

type here

Celestial Event Viewer

	Date	New Columnn	Visibility	Duration
1				
2				
3				
4				

Select row for event description

Fetch



[Go back to menu](#)
[Close](#)

Variable Stars Management (for Admin):

Type Here

Variable Stars

User

Name

Star ID

RA

Dec

Spectral Clas:

Search

Add

Delete

	Star ID	Name	Min Magnitude	Max Magnitude	Period	bservation Tim	Type
1							
2							
3							
4							
5							
6							
7							
8							

Go back to menu

Close

Stellar Data Management (for Admin):

User Management

User

Name
Star ID
RA
Dec
Spectral Clas

	Star ID	Name	ht Ascension (l	Declination(Dec	Magnitude	Spectral Class	Distance	Proper Motion
1								
2								
3								
4								
5								
6								
7								
8								

The screen for Celestial Event Viewer will be the same as the ones for Stellar Data Management and Variable Star Management. Horoscope will have the same result for users and admin.